Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/21**

Paper 2 Theory **May/June 2021**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **11** printed pages.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)(i)	E	1
1(a)(ii)	A	1
1(a)(iii)	E	1
1(a)(iv)	D	1
1(a)(v)	F	1
1(b)	C and F	1

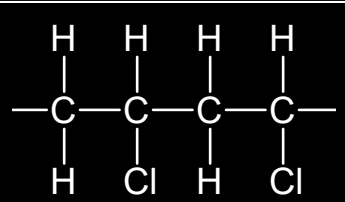
Question	Answer	Marks
2(a)	argon	1
2(b)	(air) balloons / undersea breathing apparatus	1
2(c)	outer shell has 8 electrons	1
2(d)(i)	86 protons / 86 electrons / same electronic structure	1
2(d)(ii)	radon-222 has two more neutrons / radon-220 has two fewer neutrons / radon-220 has 134 neutrons and radon-222 has 136 neutrons	1
2(e)	percentage of xenon = 60.4 (1) mole ratio Xe : O : F is 60.4 / 130 : 22.1 / 16 : 17.5 / 19 **OR** 0.46 : 1.38 : 0.92 (1) empirical formula XeO_3F_2 (1)	3

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Question	Answer	Marks
2(f)(i)	volume increases **AND** particles move faster / particles spread out more / particles move further apart	**1**
2(f)(ii)	volume decreases **AND** particles move closer together / distance between particles decreases	**1**
2(f)(iii)	moles = 21 / 24 **OR** 0.875 (1) mass = 18 (g) (1)	**2**

Question	Answer	Marks
3(a)(i)	fractional distillation	**1**
3(a)(ii)	**Any one from:** diesel – fuel for cars or lorries (1) paraffin / kerosene – heating / cooking / aviation fuel (1) bitumen – water proofing / roofing / road surfaces (1) heating oil – fuel for heating (1) lubricating or heavy oils – lubrication / fuels for ships (1)	**1**
3(a)(iii)	cracking / heating strongly (in presence of a catalyst) (1) large hydrocarbons or naphtha fraction broken down / (1)	**2**
3(b)(i)	cannot be decayed / cannot be decomposed (by natural or bacterial action)	**1**
3(b)(ii)	incineration causes poisonous gases / litter / plastic waste in the sea / killing animals / takes up space in landfill sites	**1**

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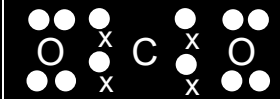
Question	Answer	Marks
3(b)(iii)	 one repeat unit shown (1) at least two repeat units and extension bonds at both ends (1)	**2**
3(b)(iv)	100 tonnes because it is addition polymerisation / it is an addition reaction / no other product is produced	**1**
3(b)(v)	CH_3CH_2Cl	**1**

Question	Answer	Marks
4(a)	**Any two from:** soluble in water (1) insoluble in organic solvents (1) high melting point / solid at room temperature (1)	**2**
4(b)	coloured / forms a blue ppt with NaOH or NH_3	**1**
4(c)	copper loses electrons and chlorine gains electrons (1) reference to (transfer of) two electrons (1)	**2**
4(d)(i)	iodide (ion) loses electrons	**1**
4(d)(ii)	Cu^{2+} (ion) gains electrons / copper(II) ions gain electrons	**1**

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Question	Answer	Marks
4(e)	copper(II) oxide **AND** carbon dioxide	**1**

Question	Answer	Marks
5(a)	**4**FeS_2 + **11**O_2 → **2**Fe_2O_3 + **8**SO_2	**1**
5(b)	acid rain / erodes buildings / corrodes buildings	**1**
5(c)	sulfur dioxide reacted with oxygen to make sulfur trioxide (1) this reaction uses a V_2O_5 catalyst / temperatures between 450 and 600 °C / 1 to 10 atmosphere pressure (1) sulfur trioxide reacted with H_2SO_4(l) to make oleum which is then reacted with water (to make sulfuric acid) (1)	**3**
5(d)	bleaching (wood pulp) / preservative	**1**
5(e)(i)	reacts with oxygen to provide energy for the reaction / makes carbon monoxide / reduces carbon dioxide	**1**
5(e)(ii)	(decomposes) to make calcium oxide (which removes impurities)	**1**
5(e)(iii)	provides oxygen for the coke to burn / makes carbon dioxide from (coke)	**1**

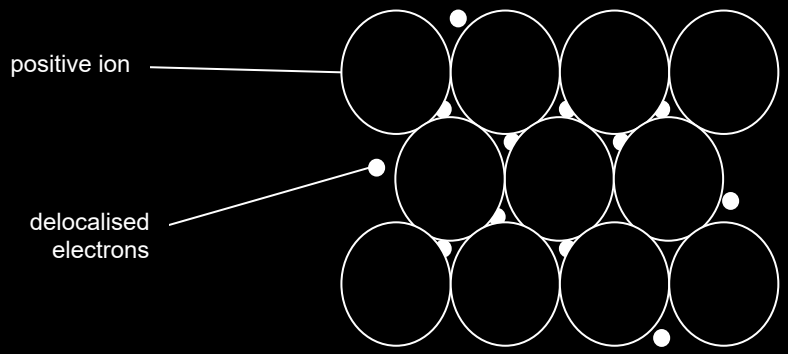
Question	Answer	Marks
6(a)(i)	methane	**1**
6(a)(ii)	global warming	**1**
6(b)		**1**
6(c)	CH_4 + $2O_2$ → CO_2 + $2H_2O$	**1**

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Question	Answer	Marks
6(d)(i)	position of equilibrium moves to the right (1) to use up the extra carbon dioxide / to reduce the concentration of carbon dioxide (1)	2
6(d)(ii)	position of equilibrium moves to the left (1) to absorb heat / because forward reaction is exothermic / backward reaction is endothermic (1)	2
6(e)(i)	little dissociation (to form ions)	1
6(e)(ii)	$Mg^{2+} + CO_3^{2-} \rightarrow MgCO_3$	1

Question	Answer	Marks
7(a)	moles of NH_4NO_2 = 0.00375 (1) volume = 0.09 (dm^3) (1)	2
7(b)	decreased (1) less crowded particles / less particles per unit volume / greater space between particles (1) collision frequency decreases / fewer collisions per second / decrease rate of collisions (1)	3
7(c)	$2NH_4NO_2 + Ca(OH)_2 \rightarrow 2NH_3 + Ca(NO_2)_2 + 2H_2O$ correct identification of ammonia as a product (1) correct formula for calcium nitrite (1) balanced equation (1)	3
7(d)(i)	(aqueous) iodine	1
7(d)(ii)	oxidation	1

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Question	Answer	Marks
8(a)	five	**1**
8(b)	 closely packed positive ions (1) delocalised electrons / sea of electrons (1) strong electrostatic attraction between the (positive) ions and the electrons (1)	**3**
8(c)	high melting point / high boiling point **AND** high density	**1**
8(d)	silver oxide / silver hydroxide **AND** nitric acid	**1**
8(e)(i)	$Ag^+ + e^- \rightarrow Ag$	**1**
8(e)(ii)	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ / $4OH^- - 4e^- \rightarrow O_2 + 2H_2O$	**1**
8(e)(iii)	ions cannot move / no mobile ions	**1**
8(f)	(light) yellow ppt.	**1**

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Question	Answer	Marks
9(a)	It has a carbon-carbon double bond	**1**
9(b)	(aqueous) bromine / Br_2 (1) orange to colourless / decolourised (1)	**2**
9(c)(i)	ethanoic acid	**1**
9(c)(ii)	lowers activation energy (1) by having an alternative pathway / alternative mechanism (1)	**2**
9(d)(i)	relative formula mass of alcohol is 58 (1) mol of alcohol = 0.2 **OR** 11.6/58 (1) mass of ester = $0.2 \times M_r$ **AND** 0.2×100 **OR** $M_r \times 11.6 / 58$ **AND** $100 \times .6 / 58$ (1)	**3**
9(d)(ii)	33.6 (%)	**1**

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